

Phase 3 Evaluation Report

Intelligent Loan Origination Agent – Retail Banking

This report documents the implementation and evaluation of the LLM-powered Loan Origination Agent developed during Phase 3 of the capstone project. The objective of this phase was to improve the baseline rule-based agent by integrating a Large Language Model (LLM), experimenting with multiple prompt strategies, and evaluating the quality, consistency, and safety of responses.

1. Phase Objectives

Integrate an LLM into the banking support workflow
Experiment with multiple prompt engineering strategies
Compare outputs across prompt variants
Analyze improvements and new failure modes
Select the most effective prompt strategy for production use

2. Updated System Architecture

Phase 2 Architecture:

User Input → Rule Engine → Static Response

Phase 3 Architecture:

User Input → Prompt Template → LLM → Structured Banking Response

3. Prompt Strategies Evaluated

Strategy 1 – Basic Prompt:

Minimal instructions with direct query forwarding.

Strategy 2 – Structured Prompt:

Provided structured response format, banking safety rules, and risk explanations.

Strategy 3 – Chain-of-Thought Prompt:

Encouraged the model to reason step-by-step before generating the final answer.

4. Evaluation Test Case

User Query:

“I earn ₹40,000 per month and have a credit score of 680. Can I get a ₹5 lakh loan?”

5. Prompt Comparison Results

Prompt Type	Output Quality	Structure	Reasoning	User Friendliness
Basic	Low	Poor	Limited	Medium
Structured	High	Excellent	Good	High
Chain-of-Thought	Medium-High	Moderate	Excellent	Medium

6. Improvements Over Baseline Agent

Natural language understanding significantly improved Dynamic responses replaced rigid rule-based outputs Improved explanation quality and user guidance Enhanced handling of ambiguous user queries Better user experience through conversational interaction

7. New Failure Modes Identified

Hallucination Risk: The LLM may generate incorrect banking rules or assumptions. **Inconsistent Responses:** Similar prompts can produce slightly different outputs. **Overconfidence:** Responses may sound certain even when information is incomplete. **Verbosity:** Chain-of-thought prompting may create overly long responses. **Safety Risk:** Weak prompts may unintentionally imply loan approval authority.

8. Selected Default Prompt Strategy

The **Structured Prompt Strategy** was selected as the default production candidate. **Justification:** Produces consistent and explainable outputs Enforces banking safety constraints Provides clear formatting for evaluation Balances reasoning quality with response clarity Improves usability for both customers and loan officers

9. Conclusion

Phase 3 successfully transformed the baseline rule-based loan assistant into an LLM-powered intelligent advisory system. The integration of prompt engineering significantly improved response quality, flexibility, and user interaction. The evaluation also revealed critical risks such as hallucination and inconsistency, highlighting the importance of safety-focused prompting and future retrieval-based grounding in subsequent phases.

Phase 3 Deliverables Completed: LLM integrated into workflow Three prompt strategies implemented Prompt comparison completed Failure analysis documented Production-ready prompt strategy selected